MicrobeTrace PrimeNG 22 Decision Assessment

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Licensing, open-source impact, alternatives, and GPT-5.6 Sol migration effort

Executive Summary

- **PrimeNG 22 solves Angular 22 compatibility but introduces a proprietary dependency.** PrimeTek's published terms exclude CDC from the Community License, and PrimeTek directly confirmed that it offers no open-source grant.
- **Buying PrimeUI is workable for deployed users but changes the development model.** Finished applications may serve unlimited users without runtime fees, while developers who build, modify, or maintain PrimeUI-based code need licensed coverage.
- **The full application becomes mixed-source.** CDC code can remain Apache 2.0, but the complete application is no longer freely buildable and modifiable under Apache terms alone.
- **AI materially changes the replacement economics.** NG-ZORRO remains the best single-suite alternative, with a likely GPT-5.6 Sol migration of 16-32 agent-active hours over 1-3 calendar days plus 2-6 hours of human review.

Use a two-track decision

Start the PrimeUI quote and clarification process, but do not let procurement decide the architecture by default. In parallel, give GPT-5.6 Sol the complete NG-ZORRO migration objective with a 2-4 hour checkpoint against the hardest table, one settings dialog, TreeSelect, reorder behavior, and targeted browser/Cypress tests.

If the checkpoint passes, continue the open-source replacement. If an Angular 22 deadline is tighter, PrimeNG 22 can be used as a documented mixed-source bridge with a reassessment or exit date. The preferred long-term outcome is the open-source replacement.

The decision is continuity versus long-term openness

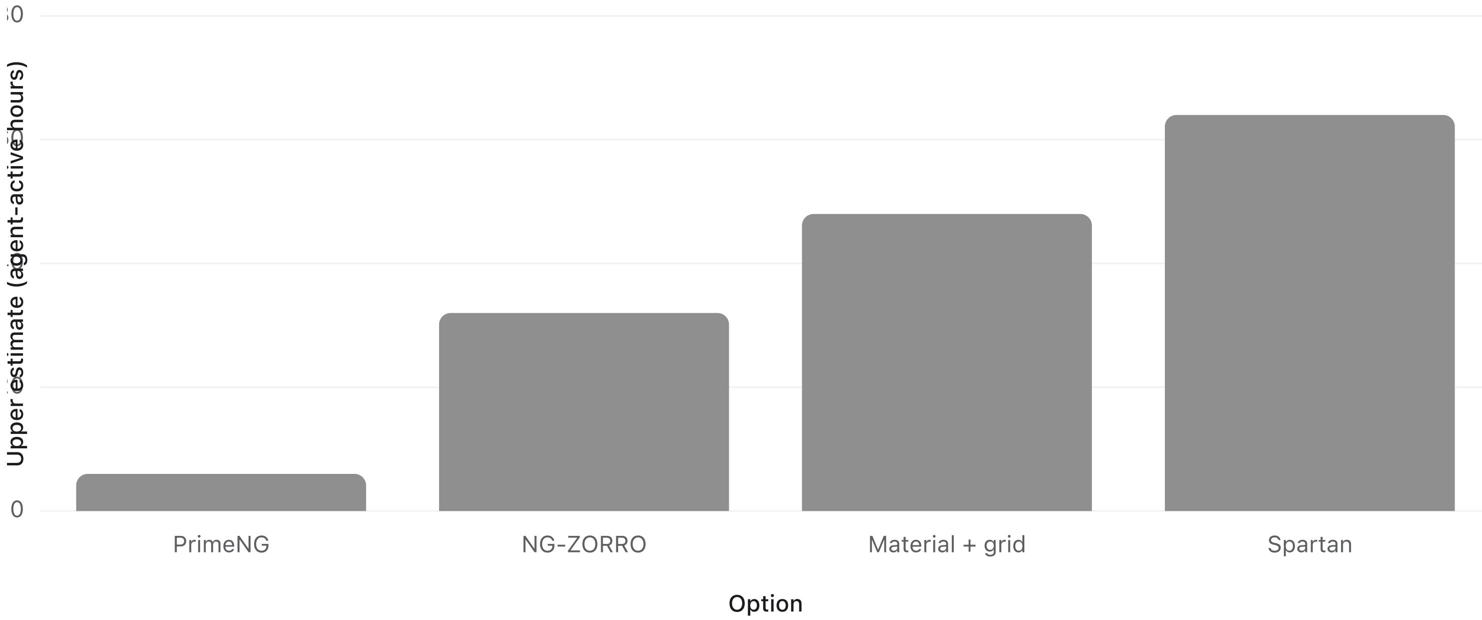
PrimeNG commercial has the lowest immediate engineering effort. NG-ZORRO has the best balance of open licensing and direct component coverage. Material/CDK with Tabulator or AG Grid Community is the governance-focused fallback. Spartan is credible but is a broader table and styling architecture change.

The effort chart compares the conservative upper bound for the four currently actionable Angular 22 options. PrimeNG's small technical bar does not include procurement elapsed time or the open-source and contributor costs; the option summary immediately afterward carries those tradeoffs and the full estimate ranges.

Estimated implementation effort by option

Upper bound of agent-active hours; procurement and the Angular 22 framework upgrade are excluded

Source: GPT-5.6 Sol option estimate



Options at a glance

- **PrimeNG 22 commercial:** Pros - lowest code churn, current UX/tests, and commercial support. Cons - proprietary dependency, procurement/renewal, and contribution friction. **2-6 technical hours plus procurement; mixed-source.**
- **NG-ZORRO 22:** Pros - MIT, Angular 22, and the closest single-suite feature coverage. Cons - different DOM/theme and test/style migration. **16-32 hours over 1-3 days; strong open-source fit.**
- **Material/CDK plus Tabulator or AG Grid Community:** Pros - MIT, Angular-team foundation, and existing Material/Tabulator use. Cons - more libraries and custom control integration. **24-48 hours over 2-5 days; strong open-source fit.**
- **Spartan plus TanStack Table:** Pros - MIT, modern Angular, accessible primitives, and style ownership. Cons - Tailwind/table architecture change and missing direct controls. **32-64 hours over 3-7 days; strong open-source fit.**
- **Angular 21 plus PrimeNG 21:** Pros - no license or immediate refactor. Cons - delayed Angular 22 and accumulating maintenance debt. **Temporary hold only.**
- **OpenNG/Optimus:** Pros - MIT fork and potentially low churn. Cons - Angular 22 remains roadmap work and governance is young. **Watchlist, not ready.**

Source: MicrobeTrace options synthesis

Commercial licensing is viable, but not equivalent to open source

Current PrimeUI terms list $599 per active developer through December 31, 2026, $799 beginning in 2027, and $399 per developer per year to extend update access. Covered versions may be used perpetually. Seats may be reassigned but may not cover concurrent developers. CI/CD systems do not require machine seats, and finished applications may be distributed to unlimited users without runtime fees.

PrimeTek states that PrimeNG 22 is compiled commercial software, government organizations are not eligible for the Community License, and no separate open-source program exists. Its email to the MicrobeTrace team confirms that no open-source grant is available. Pure code-review comments are not expressly classified in the published summary; external contributors who submit code changes are much more clearly within the licensed-developer definition.

User partnerships are mostly insulated; contributor partnerships are not

The licensing trigger is development with PrimeUI, not ordinary use of the finished application. This supports the team's view that user-side partnerships and bioinformatics integrations are unlikely to be disrupted, assuming partners are not modifying or rebuilding the PrimeUI-based interface. The material impact falls on CDC developers, contractors, STLT developers, forks, and community contribution workflows.

- **Low impact:** public-health users, user-side STLT partners, and bioinformatics integrations that consume the deployed app, files, or APIs.
- **Direct impact:** CDC developers and contractors modifying PrimeUI-based code; each concurrent developer needs licensed coverage.
- **Material impact:** STLT or community developers who clone, build, fork, or contribute to the complete UI.
- **Low technical impact but clarification needed:** public CI/CD requires no machine seat, but public key/package/build handling should be confirmed.
- **Material classification impact:** CDC code remains Apache 2.0 while the complete dependency stack becomes mixed-source.

Source: PrimeUI license and CDC partner-impact synthesis

The migration is broad but mechanically tractable

The repository contains repeated component markup, imports, styles, and test selectors that a coding agent can change in compile-driven batches. The eight tables and overlay/focus behavior are the real uncertainty because they require behavioral judgment and repeated browser/test verification. Angular Material and Tabulator are already installed, which reduces the setup cost of a hybrid fallback.

Source: MicrobeTrace repository static scan, July 31, 2026

- **Component markup:** 202 active top-level PrimeNG tags across 16 templates.
- **Imports and themes:** 16 TypeScript files import PrimeNG or its theme package.
- **Application styles:** 13 files depend on PrimeNG implementation classes.
- **Cypress/support:** about 573 PrimeNG DOM/class references across 96 maintained files.
- **Highest risk:** eight tables with filtering, selection, paging, virtual scroll, expansion, export, reorder, and direct API use.

Source: MicrobeTrace repository static scan, July 31, 2026

NG-ZORRO is the first replacement to try

NG-ZORRO 22 is MIT licensed, supports Angular 22, and has the closest direct coverage for MicrobeTrace's table, TreeSelect, modal, select, collapse, and transfer needs. Its main costs are DOM/test changes and adopting or restyling Ant Design.

Material/CDK plus Tabulator is the strongest consolidation fallback because both are already present. AG Grid Community is worth testing only if Tabulator cannot reproduce the critical table behaviors without paid features.

Spartan is a sound modern design-system option and is actively compatible with Angular 22. It is not the shortest path here: its data table is a TanStack recipe, and several higher-risk PrimeNG controls need custom composition.

OpenNG/Optimus may eventually be the lowest-churn open fork, but Angular 22 support remains roadmap work and the project is still maturing.

Sol can likely produce a tested candidate in 1-3 days

The estimate assumes one GPT-5.6 Sol coding agent with repository write access, dependency installation, the local app, Playwright, and Cypress. It is an engineering estimate rather than an OpenAI speed guarantee and carries approximately plus or minus 50% uncertainty until the hard checkpoint is complete. The likely range is 16-32 agent-active hours, with 2-6 hours of human review for approvals, visual judgment, legal/program decisions, and final diff acceptance.

- **Foundation and adapters (1-3 hours):** dependencies, application-owned types, theme shell, and stable test hooks.
- **Repeated controls and overlays (3-6 hours):** selects, dialogs, accordions, and simple controls in compile-driven batches.
- **TreeSelect, reorder, and tables (5-10 hours):** direct component APIs and highest-risk view behavior.
- **Cypress decoupling and repair (3-6 hours):** replace PrimeNG DOM coupling with application-owned selectors.
- **Browser, accessibility, and regression verification (4-12 hours):** Playwright evidence, maintained Cypress suites, visual checks, and fixes.

Phase maxima should not be added mechanically because compile, browser, and test repair overlap.

Source: GPT-5.6 Sol migration phase estimate

Recommended next steps

1. Request a PrimeUI quote and determine the number of concurrent CDC and contractor developer seats.
2. Ask PrimeTek in writing about public repositories, forks, external pull requests, public CI builds, offline distribution, and license-key handling.
3. Run the 2-4 hour NG-ZORRO checkpoint on the existing replacement branch.
4. Continue the migration if the hard behaviors pass; otherwise preserve the evidence and use PrimeNG 22 as a time-boxed mixed-source bridge if the release date requires it.
5. If PrimeNG 22 is chosen, update the repository's licensing language, dependency notices, build instructions, SBOM, and contributor guidance, and set a reassessment date.

Questions the program still needs to close

- What is the required Angular 22 release date, and does it allow a 1-3 day replacement attempt?
- Will the program accept a mixed-source description and a public repository that is not independently buildable under Apache terms alone?
- How many concurrent developer seats are required, including contractors?
- What written contribution and build terms will PrimeTek provide?
- What exit trigger prevents a bridge license from becoming an indefinite procurement dependency?

Caveats and assumptions

This is an engineering and program analysis, not legal advice. CDC legal/open-source and acquisition reviewers should confirm final classification, redistribution, procurement, and license interpretation. Partner impact is inferred from PrimeTek's published distinction between finished-application users and developers; a partner that modifies or rebuilds the UI would fall on the developer side of that distinction. Effort estimates exclude the Angular 22 framework upgrade and procurement elapsed time.

Reference links

- PrimeUI licensing announcement
- PrimeUI Commercial License
- PrimeUI Community License
- PrimeUI pricing
- NG-ZORRO Angular 22 and MIT support
- Spartan Angular support and data-table implementation
- OpenNG/Optimus roadmap
- PrimeTek email and CDC program discussion supplied to the MicrobeTrace team, July 2026